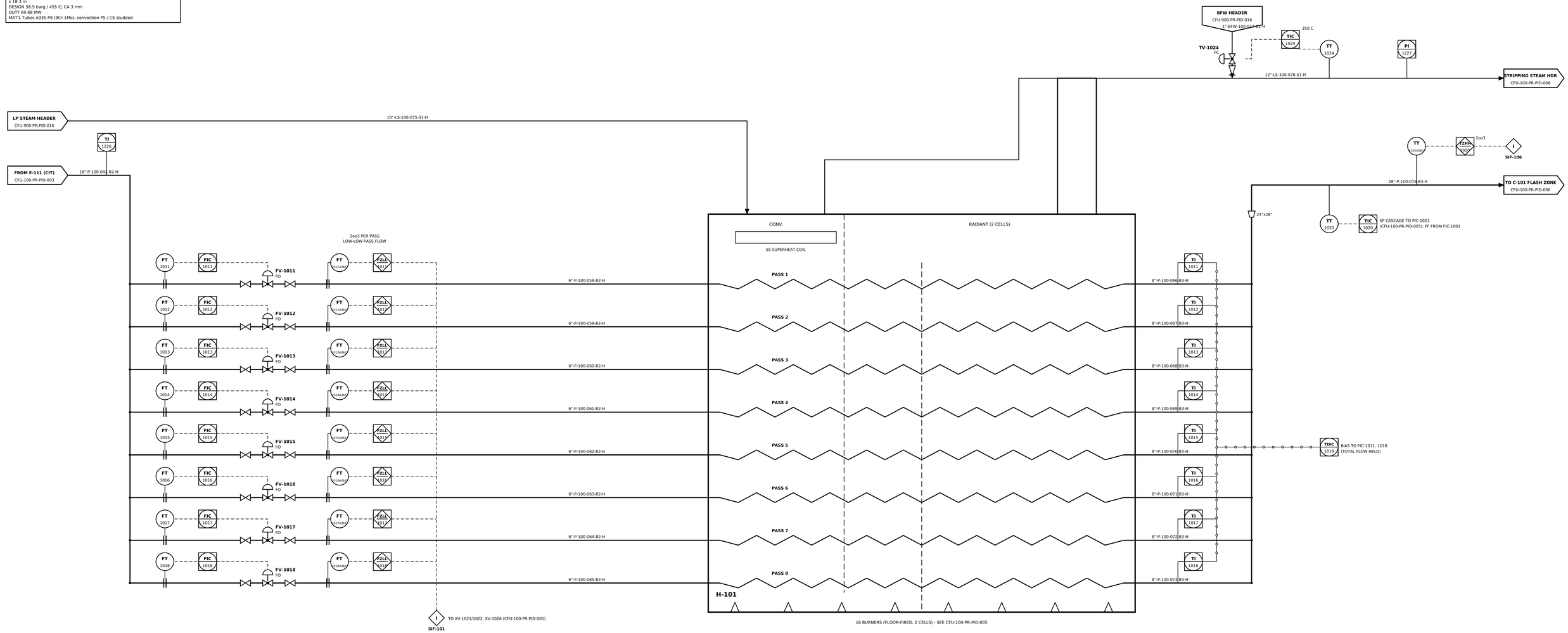
H-101

Atmospheric crude charge heater
60.7 MW abs. / 67.4 MW fired; 2-cell cabin, 8 passes, 128 rad. tubes 6" x 18.3 m
DESIGN 38.5 barg / 455 C; CA 3 mm
DUTY 60.68 MW
MAT'L Tubes A335 P9 (9Cr-1Mo); convection P5 / CS studded



NOTES

1. Passes shown schematically (convection + radiant in series per pass); 8 identical passes.
2. Tube-skin thermocouples: 4 per pass (TI-1101..1132) - shown on heater vendor P&ID, alarmed in DCS.
3. Pass flow SIS transmitters FT-10xxA/B/C (2oo3 per pass) independent from BPCS FT-10xx (SIF-101).
4. Decoking (steam-air) connections and spool pieces at inlet / outlet manifolds by heater vendor.
5. Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
6. All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
7. Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - *GULF COAST*
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PIPING & INSTRUMENT DIAGRAM

ATMOSPHERIC HEATER H-101 - PROCESS COILS

DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	5 OF 17

DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02

DRAWING NUMBER: **CFU-100-PR-PID-004**

REV: **A**